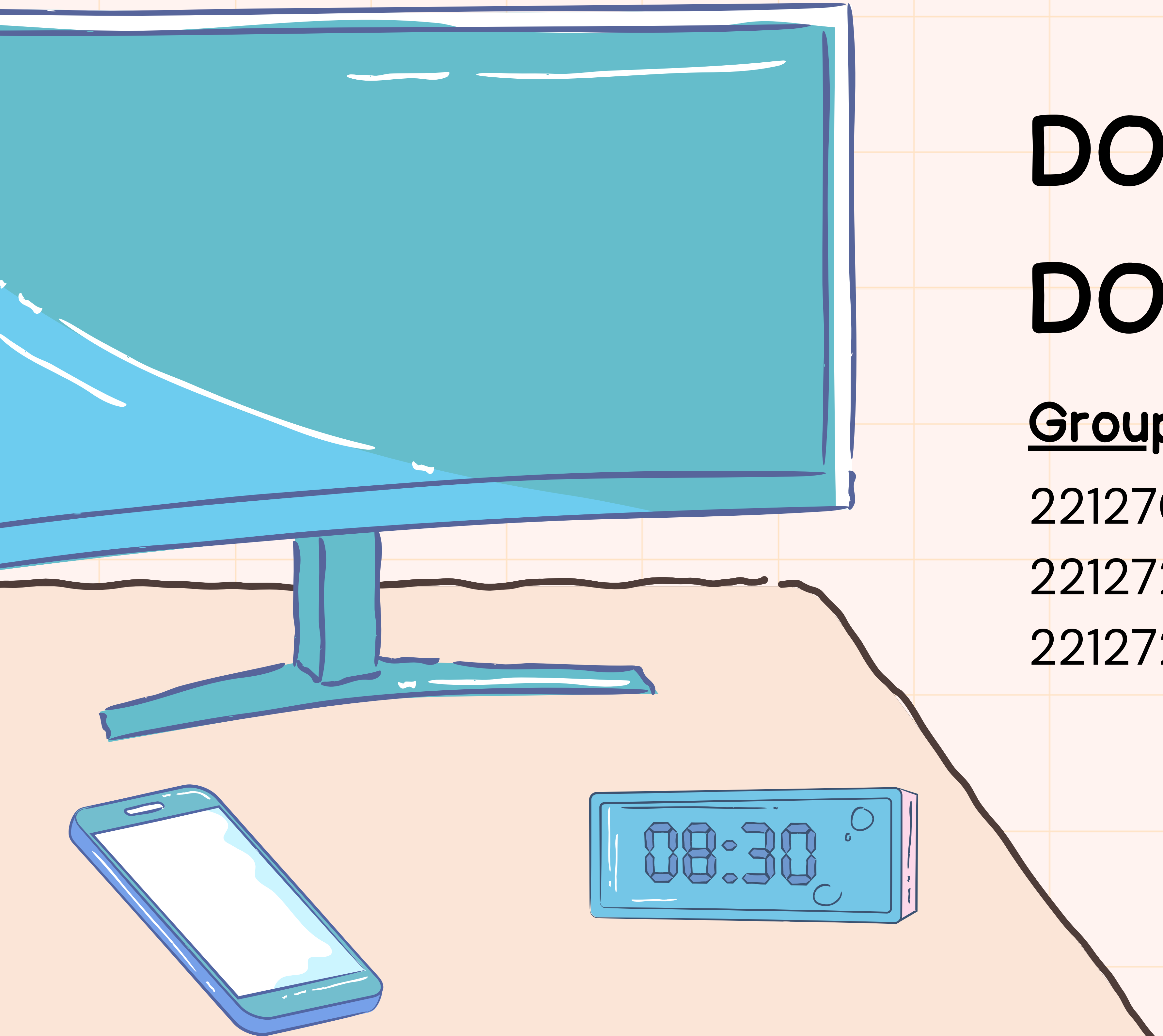

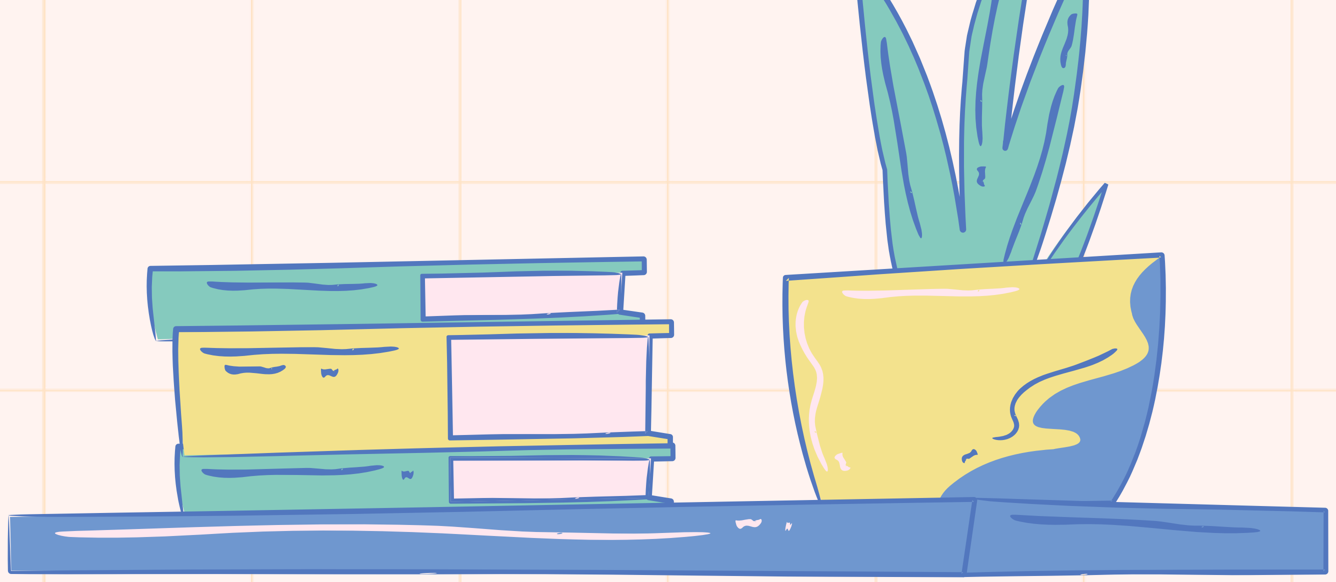

DOCBERT: BERT FOR DOCUMENT CLASSIFICATION

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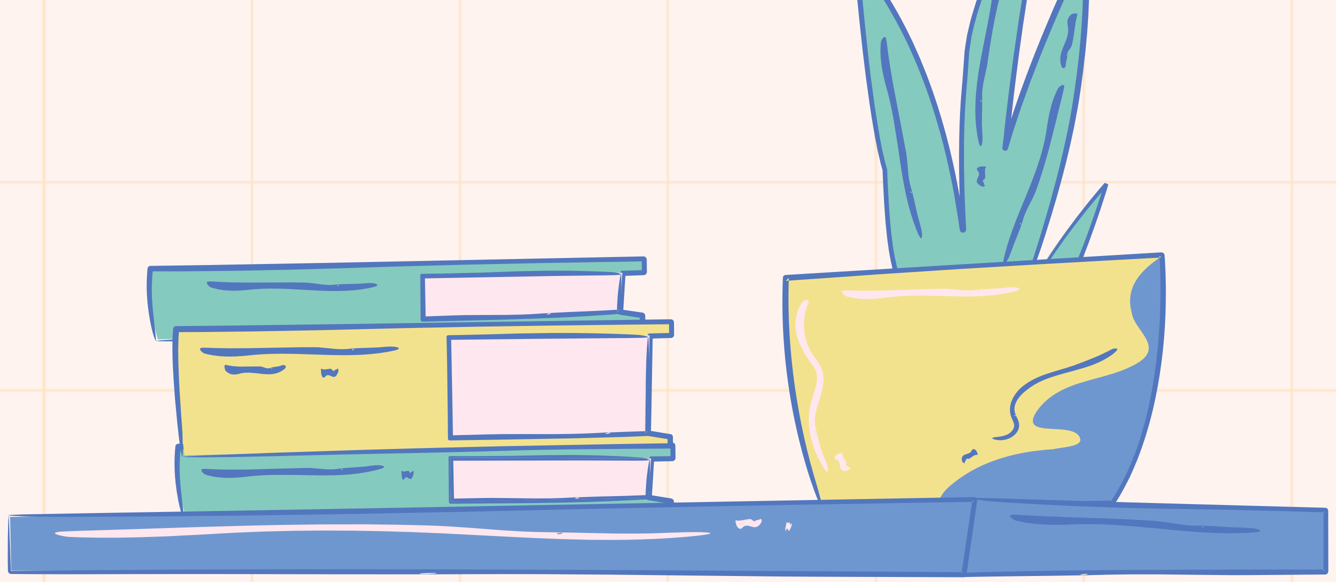
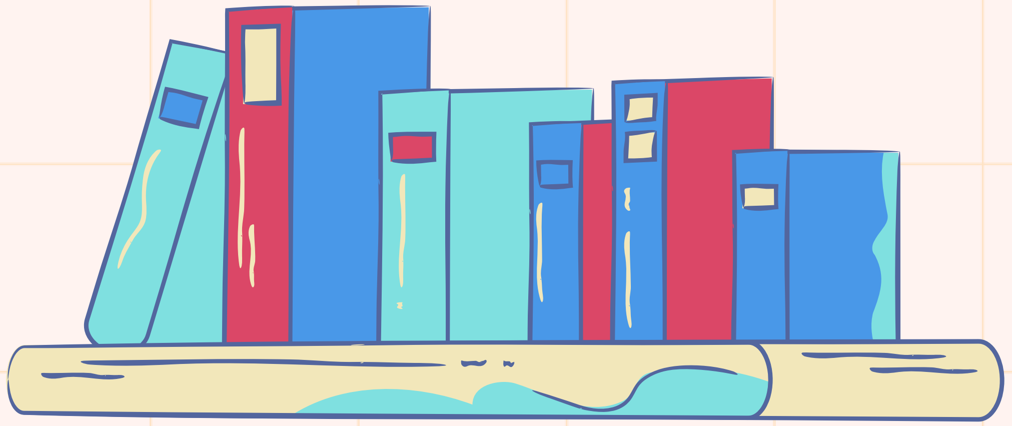
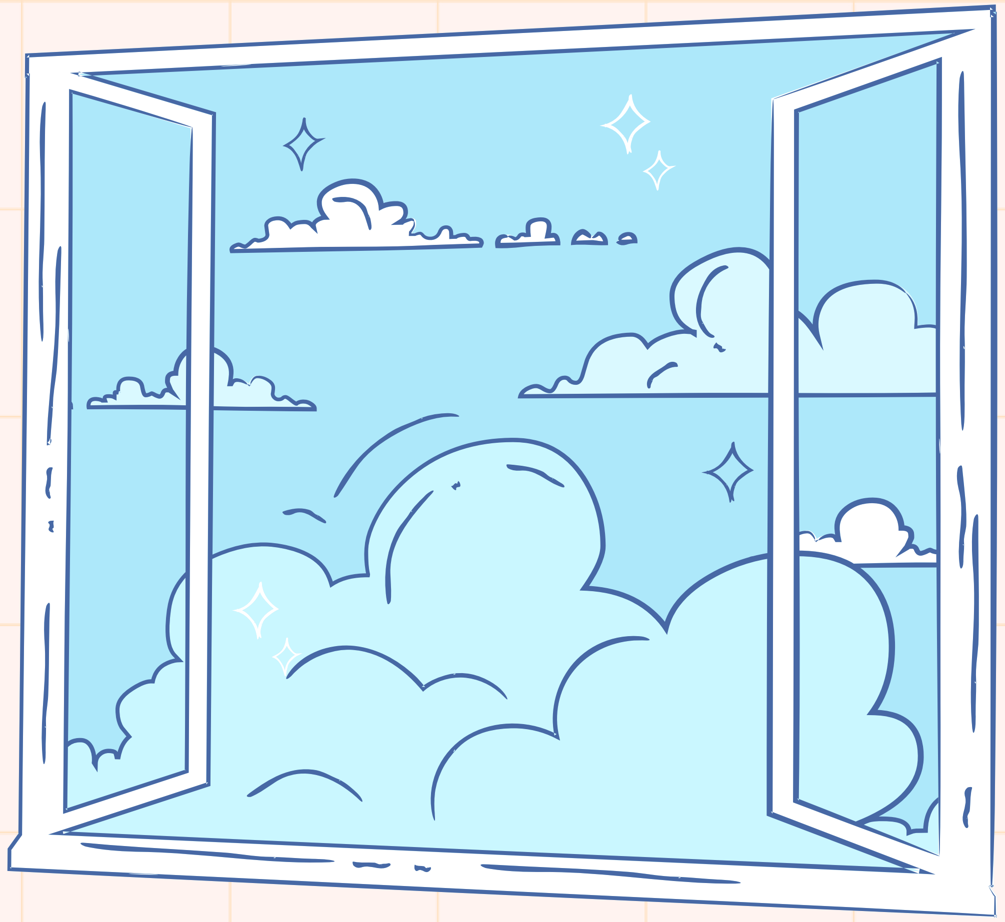


OUTLINE

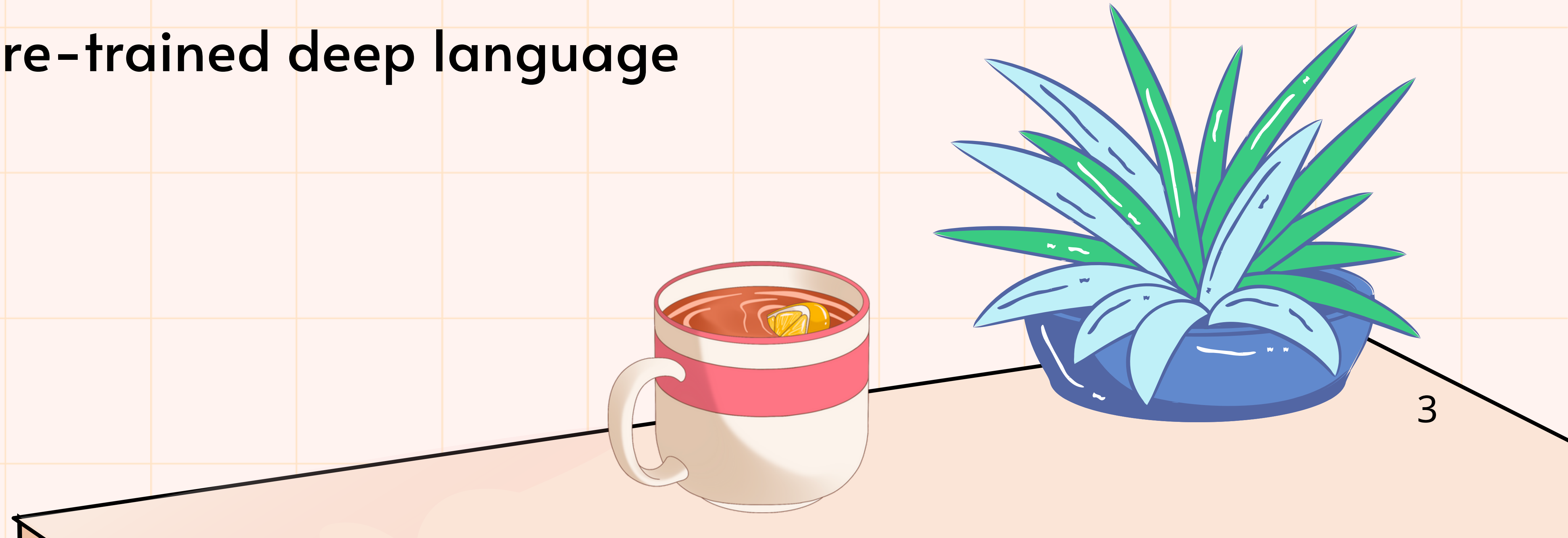
1. Introduction
2. Approaches
3. Experimental process
4. Result
5. Conclusion

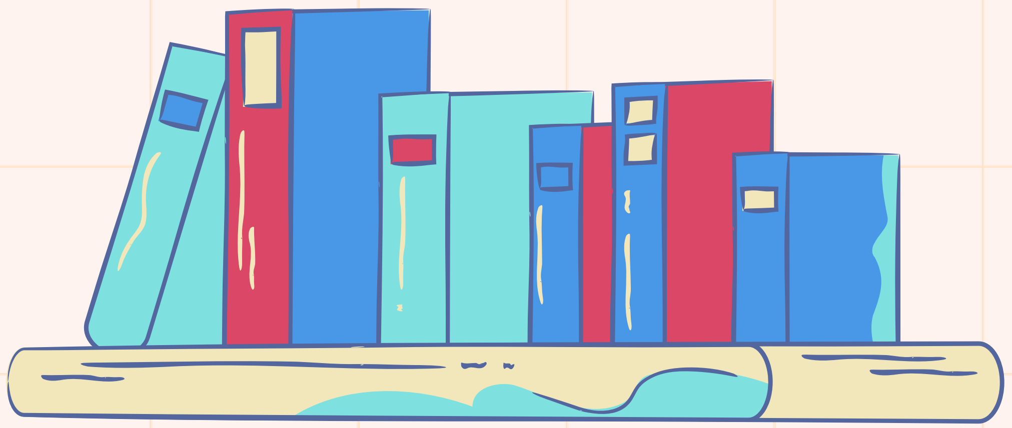


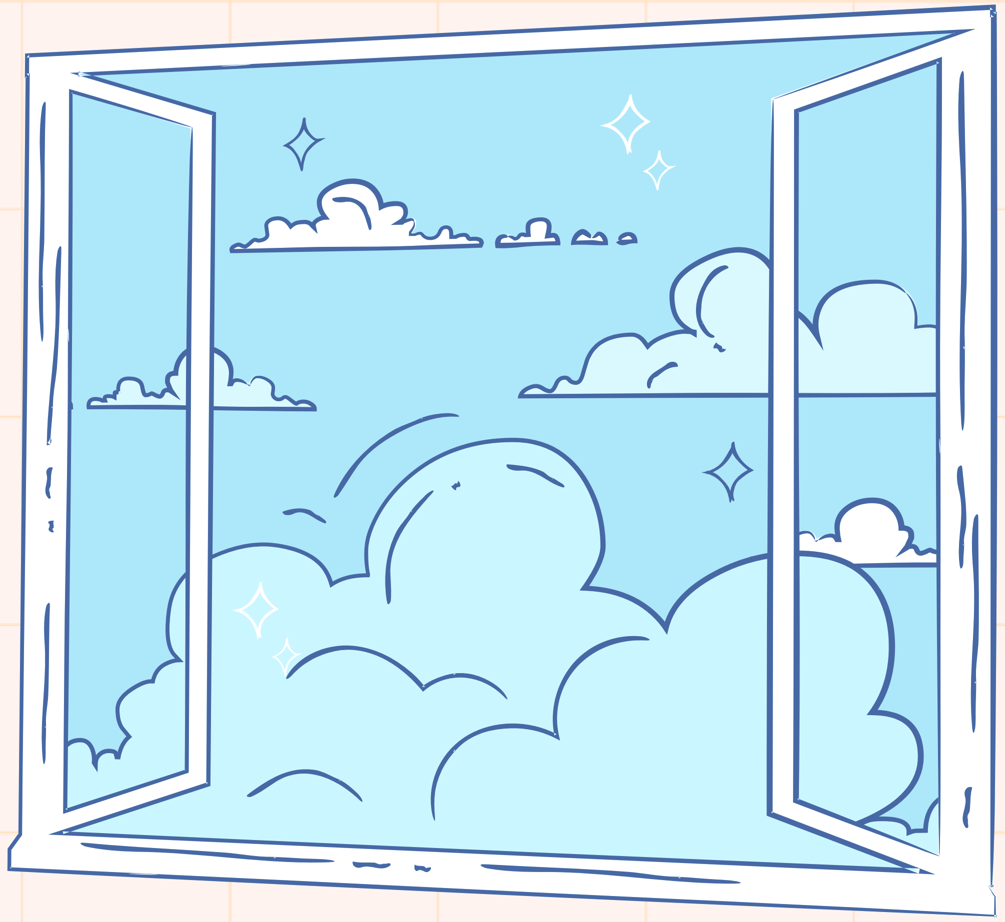


1. Introduction

- Document classification is a very important task in NLP
- Until recently, the dominant model in approaching natural language processing (NLP) tasks has been to concentrate on neural architecture design
- Currently, NLP is shifting toward pre-trained deep language representation models.

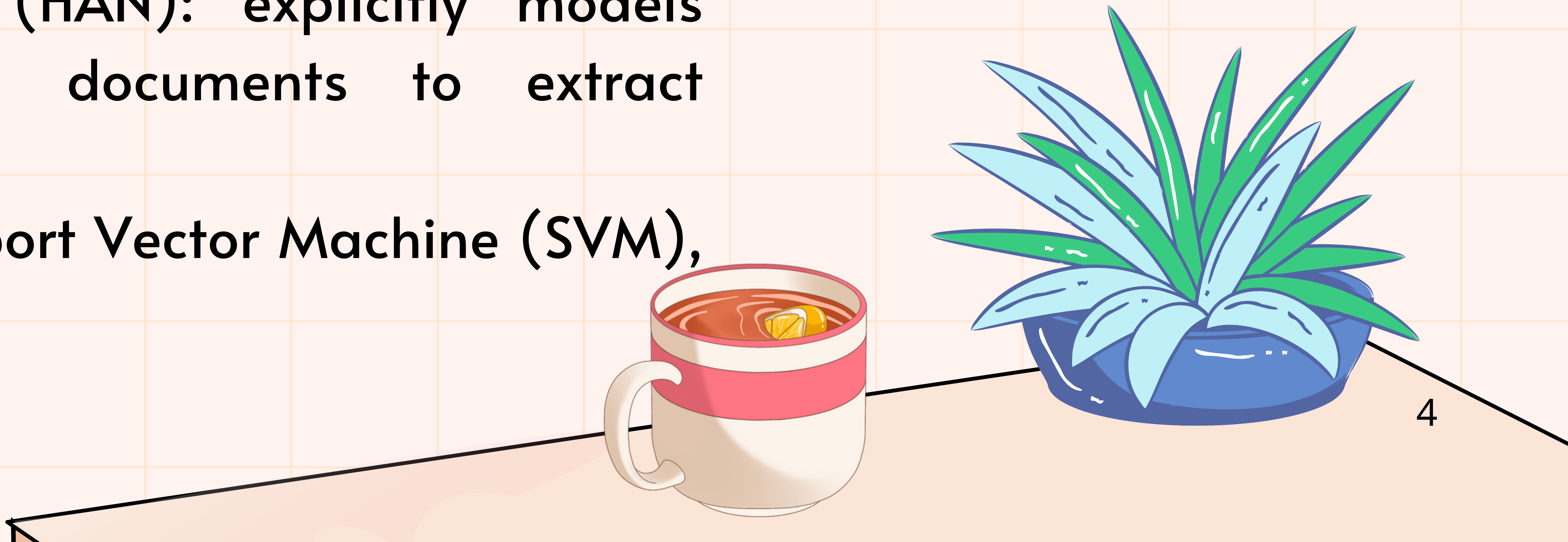






Some common predecessor models :

- LSTMreg: a simple single-layer BiLSTM model.
- XML-CNN: designed to address the multi-label nature of the problem.
- Hierarchical Attention Network (HAN): explicitly models hierarchical information within documents to extract meaningful features.
- Logistic Regression (LR) and Support Vector Machine (SVM), trained on tf–idf vectors.



Some issues

- Overly long documents: BERT is limited to 512 tokens. Truncating documents may harm model quality, especially on datasets with long texts such as IMDB.
- Multi-label Nature: Documents often have multiple labels across many classes. This characteristic is not typical of the traditional tasks BERT is designed for.
- Less dependence on Syntactic Structure: Modeling syntactic structure has been arguably less important for document classification compared to typical BERT tasks, such as natural language inference and paraphrasing.

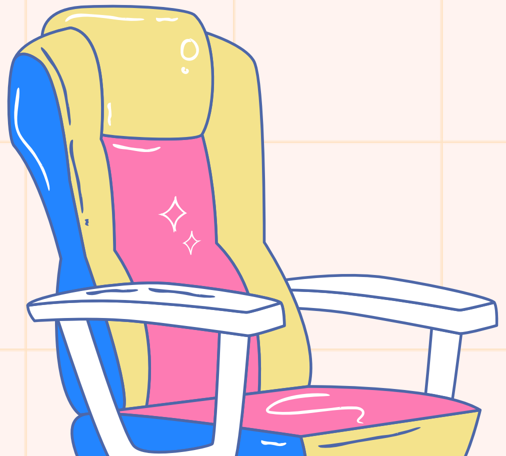
Solutions

A) Increase in Model Quality

Using BERT (DocBERT): utilizing a large language model that has been pre-trained on vast amounts of text.

Approach:

1. Pre-training: BERT is pre-trained to understand language.
2. Fine-tuning: Next, this pre-trained network is then fine-tuned on task-specific, labeled data (document classification data).



Solutions

B) Knowledge Distillation - KD

Apply Knowledge Distillation to transfer knowledge learned by a larger model (the teacher) to a smaller model (the student). Specifically, KD is applied to transfer knowledge from BERTlarge (the large BERT variant) to a much smaller Bidirectional LSTM (BiLSTM)

- <u>Benefit</u>: The smaller model can achieve accuracy close to the larger model while running significantly faster.
- <u>Training objective</u>: The model is trained both to correctly classify the original labels and to mimic the probability distribution produced by the larger model (the 'soft targets')

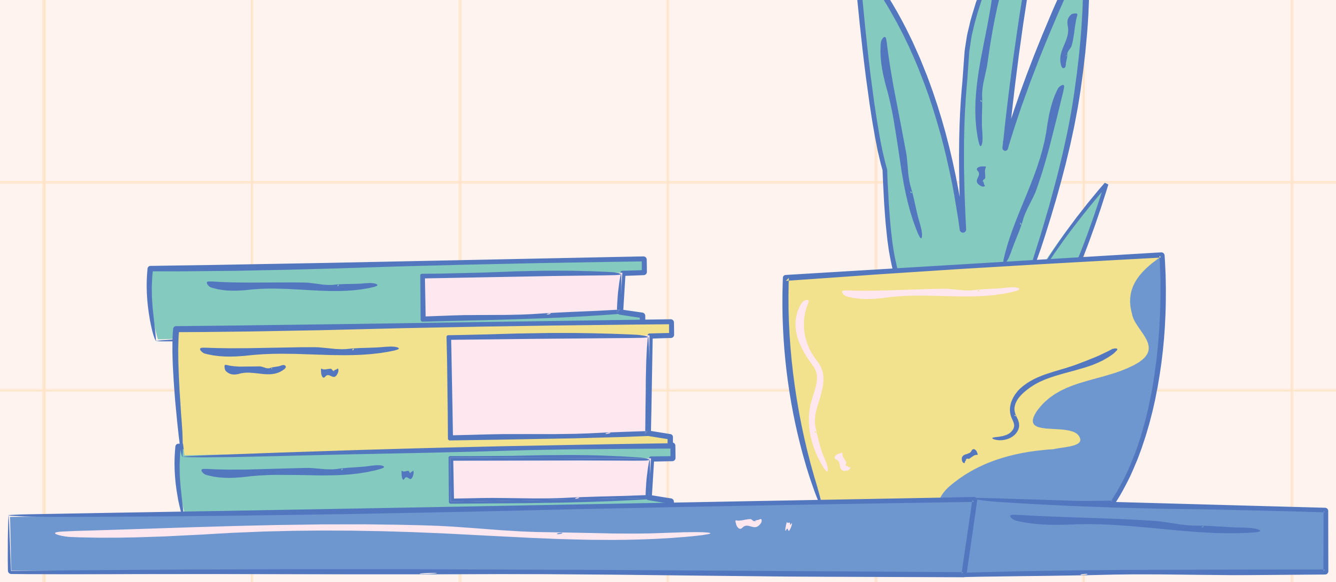
Result

A. Model Quality

- SOTA: Simply fine-tuning the BERT-large model achieves state-of-the-art results on all four standard document classification datasets.
- Performance: The student model after knowledge distillation (KD-LSTMreg) attains performance comparable to the BERT-base model.

B. Computational Efficiency

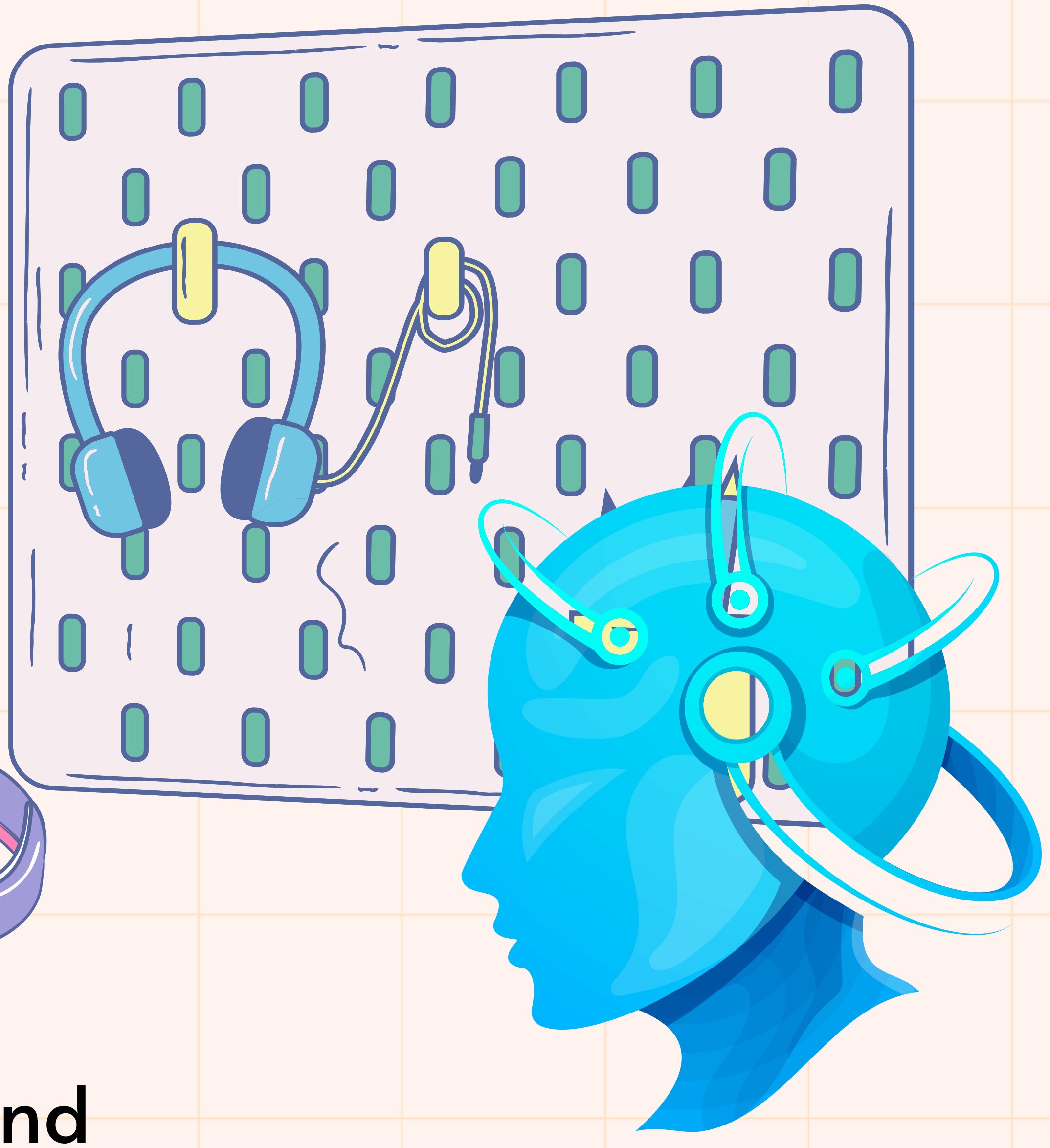
- Parameter reduction: The KD-LSTMreg model reduces the number of parameters by up to 30 times compared to BERT-base.
- Faster inference speed: The KD-LSTMreg model performs inference at least 40 times faster than BERT-base.



2. APPROACHES

BERT

- Uses Transformer architecture
- Learns contextual representations from both left and right



Pre-training objectives:

- Masked Language Modeling
- Next Sentence Prediction

Captures rich contextual information, including sentence content and inter-sentence relationships
After pre-training, it is highly flexible and can be fine-tuned for many NLP tasks

BERT Fine-tunning

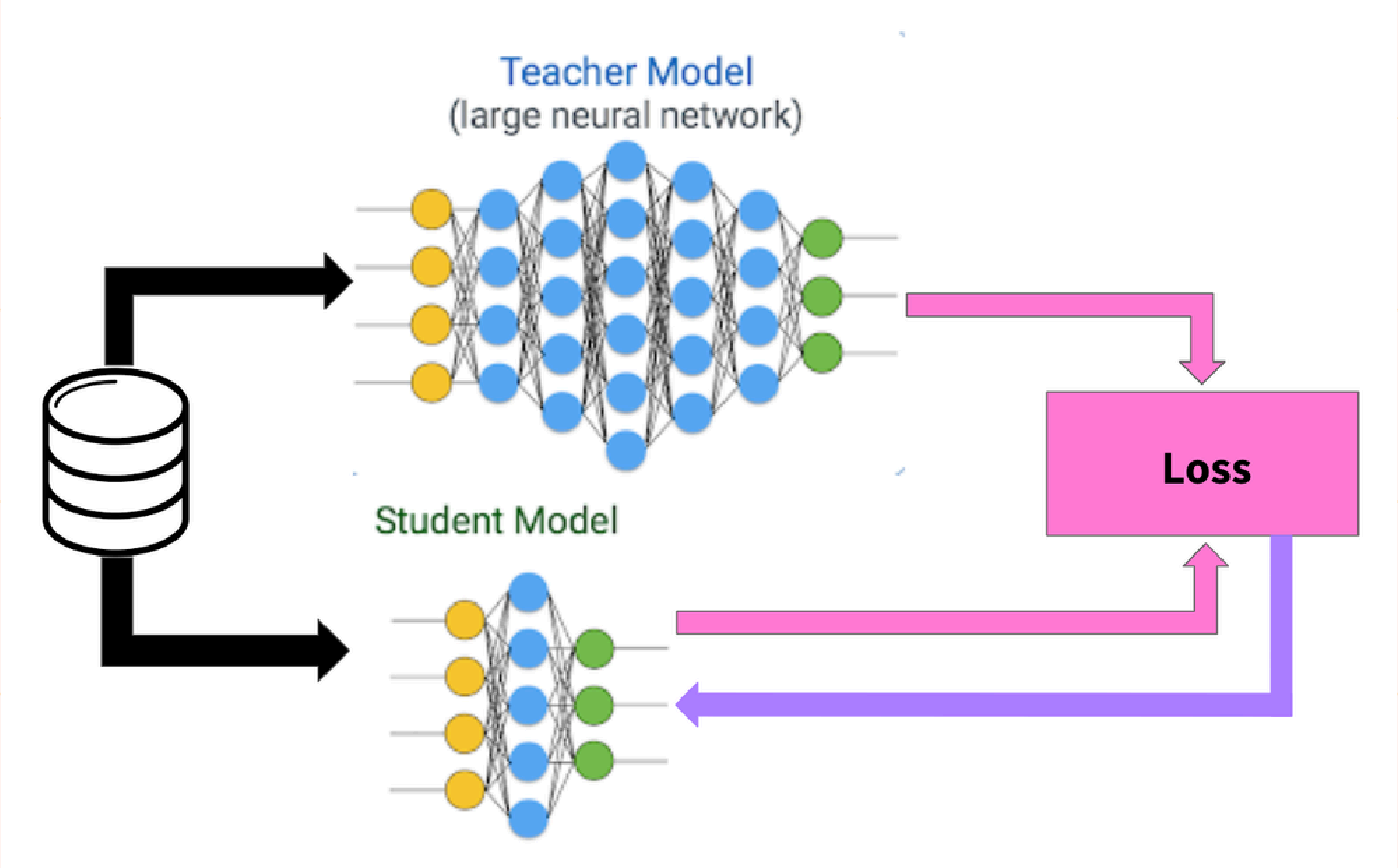
- Fine-tune BERT (BERT-base and BERT-large) for document classification by adding a softmax (fully connected) layer on top of the [CLS] token.
- The entire model is fine-tuned end-to-end together with the softmax classifier:

$$W \in \mathbb{R}^{K \times H}$$

- Loss Function:
    - Cross-entropy for single-label classification
    - Binary cross-entropy for multi-label classification

Knowledge Distillation

- A model compression technique where the knowledge of a large model (teacher) is “distilled” and transferred to a smaller model (student).
- The student is trained on the transfer set using both the target labels and the soft targets the probability distribution produced by the pre-trained teacher model.
- It is not dependent on the model architecture, allowing knowledge to be transferred between very different types of models.



KD-LSTM

- Compress the BERT model (teacher) into a smaller model (student), specifically a Bi-LSTM (LSTMreg).
- Transfer set: The training set is augmented using data augmentation techniques:
  - POS-guided word swapping
  - Random masking

→ The student is trained by combining the target labels with the outputs from the teacher model:

$$\mathcal{L} = \mathcal{L}_{classification} + \lambda \cdot \mathcal{L}_{distill}$$

KD-LSTM

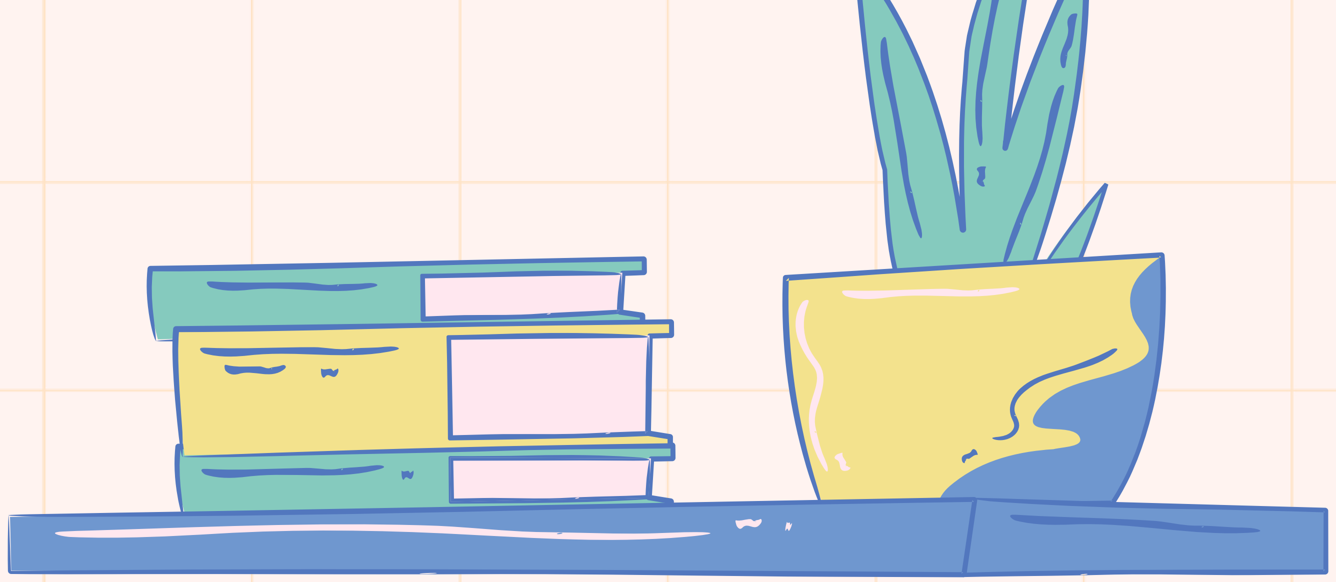
$$\mathcal{L} = \mathcal{L}_{classification} + \lambda \cdot \mathcal{L}_{distill}$$

$\mathcal{L}_{classification}$: based on the target labels, similar to standard BERT training.

$\mathcal{L}_{distill}$: Kullback–Leibler (KL) divergence

$$KL(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i}$$

p: predicted probability distribution from the student model
q: probability distribution produced by the teacher model (BERT) for the same input sample

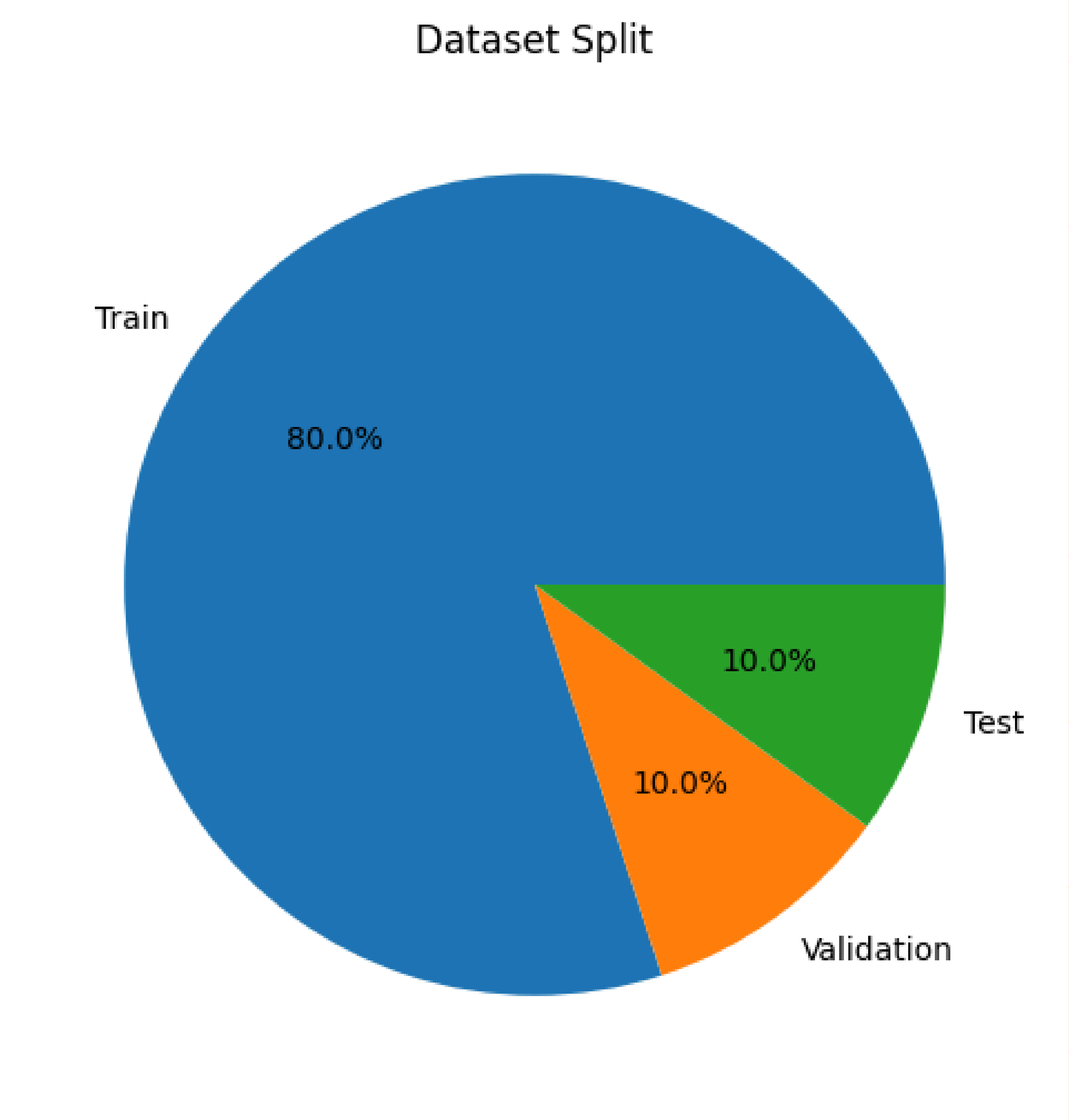


3. EXPERIMENTAL PROCESS

Datasets

Dataset	C	N	W	S
Reuters	90	10,789	144.3	6.6
AAPD	54	55,840	167.3	1.0
IMDB	10	135,669	393.8	14.4
Yelp 2014	5	1,125,386	148.8	9.1

Table 1: Summary of the datasets. C denotes the number of classes in the dataset, N the number of samples, and W and S the average number of words and sentences per document, respectively.

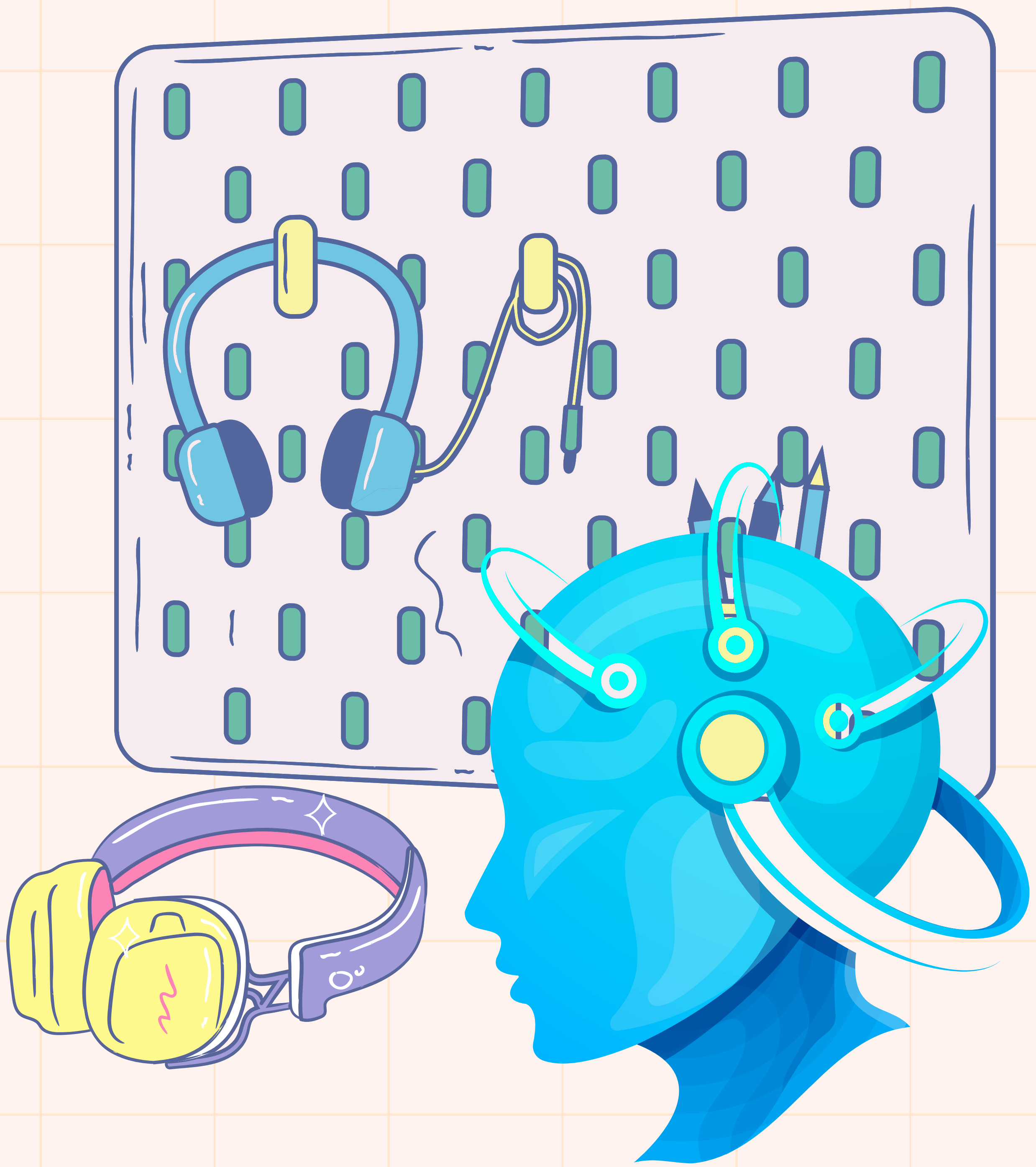


Experimental Setup

- Model comparison:
    - Fine-tuned BERT vs. HAN, KimCNN, XML-CNN, SGM, LSTMreg
- Hardware:
    - Fine-tuning BERT: Nvidia Tesla V100, Tesla P100
    - Other experiments: RTX 2080 Ti, GTX 1080
- Frameworks & tools:
    - PyTorch 0.4.1 (training)
    - Scikit-learn 0.19.2 (tf–idf, Logistic Regression, SVM)

Fine-tunning Bert

- Optimize the following hyperparameters:
    - Epochs
    - Batch size
    - Learning rate
    - Maximum Sequence Length (MSL) the maximum number of tokens before truncation
- The model is sensitive to the number of epochs, so this needs to be tuned for each dataset



Fine-tunning Bert

- Number of epochs for each dataset:
  - Reuters: 30 epochs
  - AAPD: 20 epochs
  - IMDB: 4 epochs
  - Yelp 2014: 1 epoch (due to resource constraints)
- Optimized hyperparameters:
  - Batch size: 16
  - Learning rate: 2×10^5
  - MSL: 512 tokens

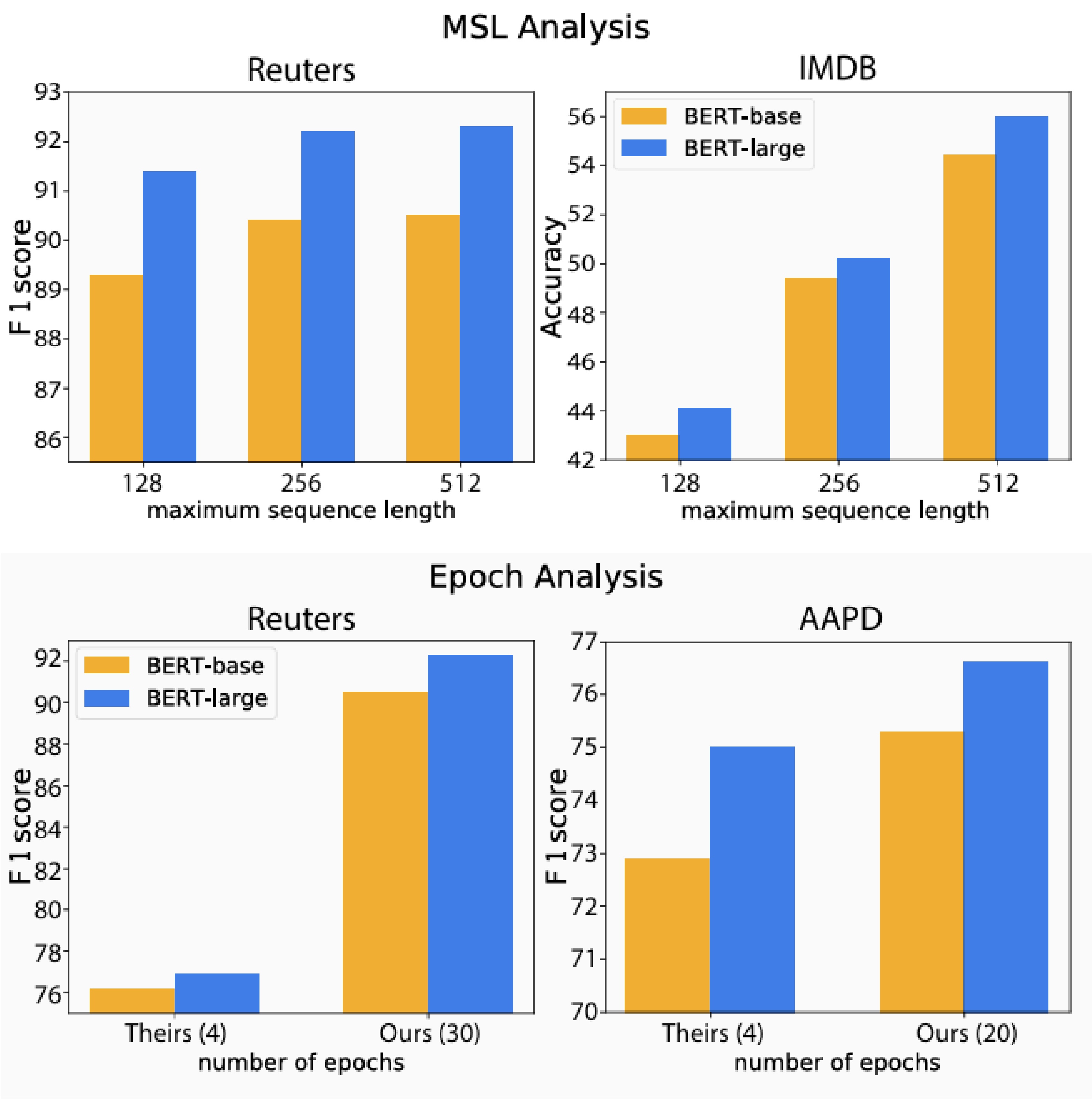


Figure 2: Results on the validation set from varying the MSL and the number of epochs.

Distillation Process

- Model:

LSTM_reg is trained to learn the representations distilled from BERT_large, using the objective shown in Equation:

$$\mathcal{L} = \mathcal{L}_{classification} + \lambda \cdot \mathcal{L}_{distill}$$

- Batch size:
    - 128 for multi-label tasks
    - 64 for single-label tasks

Transfer Set

- Data augmentation:
  - POS-guided word swapping
  - Random masking
- Sizes:
  - Reuters: ×3
  - IMDB: ×4
  - AAPD: ×4
  - Yelp 2014: ×1 (no data augmentation due to lacking of resource)
- Coefficient λ in the objective function:
  - Multi-label: λ = 1
  - Single-label: λ = 4

$$\mathcal{L} = \mathcal{L}_{classification} + \lambda \cdot \mathcal{L}_{distill}$$

4.RESULT

#	Model	**Reuters**		**AAPD**		**IMDB**		**Yelp ’14**	
		Val. F_1	Test F_1	Val. F_1	Test F_1	Val. Acc.	Test Acc.	Val. Acc.	Test Acc.
1	LR	77.0	74.8	67.1	64.9	43.1	43.4	61.1	60.9
2	SVM	89.1	86.1	71.1	69.1	42.5	42.4	59.7	59.6
3	KimCNN Repl.	83.5 ±0.4	80.8 ±0.3	54.5 ±1.4	51.4 ±1.3	42.9 ±0.3	42.7 ±0.4	66.5 ±0.1	66.1 ±0.6
4	KimCNN Orig.	–	–	–	–	–	37.6[8]	–	61.0[8]
5	XML-CNN Repl.	88.8 ±0.5	86.2 ±0.3	70.2 ±0.7	68.7 ±0.4	–	–	–	–
6	HAN Repl.	87.6 ±0.5	85.2 ±0.6	70.2 ±0.2	68.0 ±0.6	51.8 ±0.3	51.2 ±0.3	68.2 ±0.1	67.9 ±0.1
7	HAN Orig.	–	–	–	–	–	49.4[3]	–	70.5[3]
8	SGM Orig.	82.5 ±0.4	78.8 ±0.9	–	71.0[2]	–	–	–	–
9	LSTM_{reg}	89.1 ±0.8	87.0 ±0.5	73.1 ±0.4	70.5 ±0.5	53.4 ±0.2	52.8 ±0.3	69.0 ±0.1	68.7 ±0.1
10	BERT_{base}	90.5	89.0	75.3	73.4	54.4	54.2	72.1	72.0
11	BERT_{large}	**92.3**	**90.7**	**76.6**	**75.2**	**56.0**	**55.6**	**72.6**	**72.5**
12	$\mathrm{KD\text{-}LSTM}_{reg}$	91.0 ±0.2	88.9 ±0.2	75.4 ±0.2	72.9 ±0.3	54.5 ±0.1	53.7 ±0.3	69.7 ±0.1	69.4 ±0.1

Table 2: Results for each model on the validation and test sets. Best values are bolded. *Repl.* reports the mean of five runs from our reimplementations; *Orig.* refers to point estimates from †Yang et al. (2018), ‡Yang et al. (2016), and ††Tang et al. (2015). $\mathrm{KD\text{-}LSTM}_{reg}$ represents the distilled LSTM_{reg} using the fine-tuned BERT_{large}.

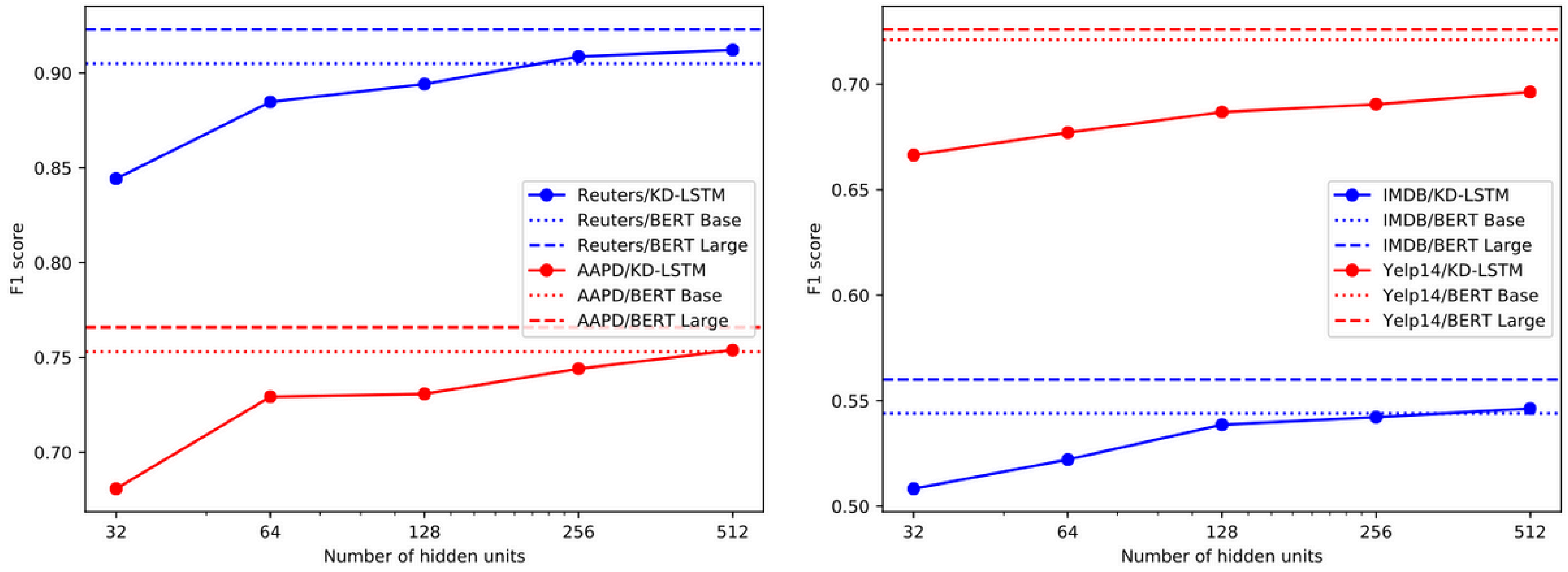
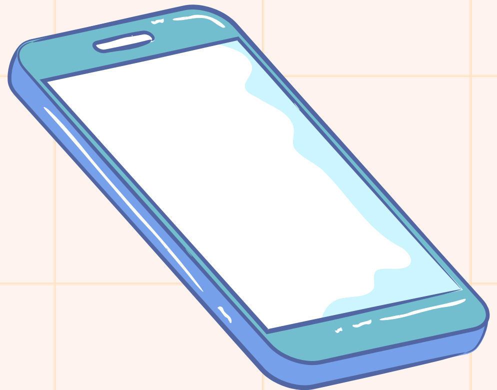
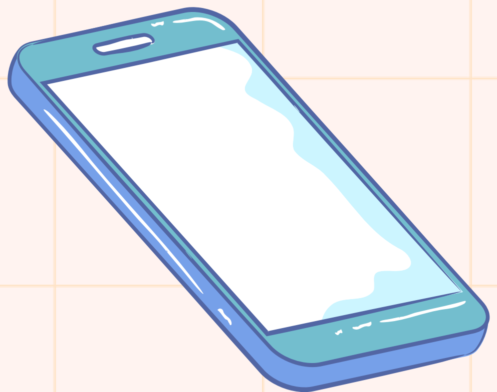


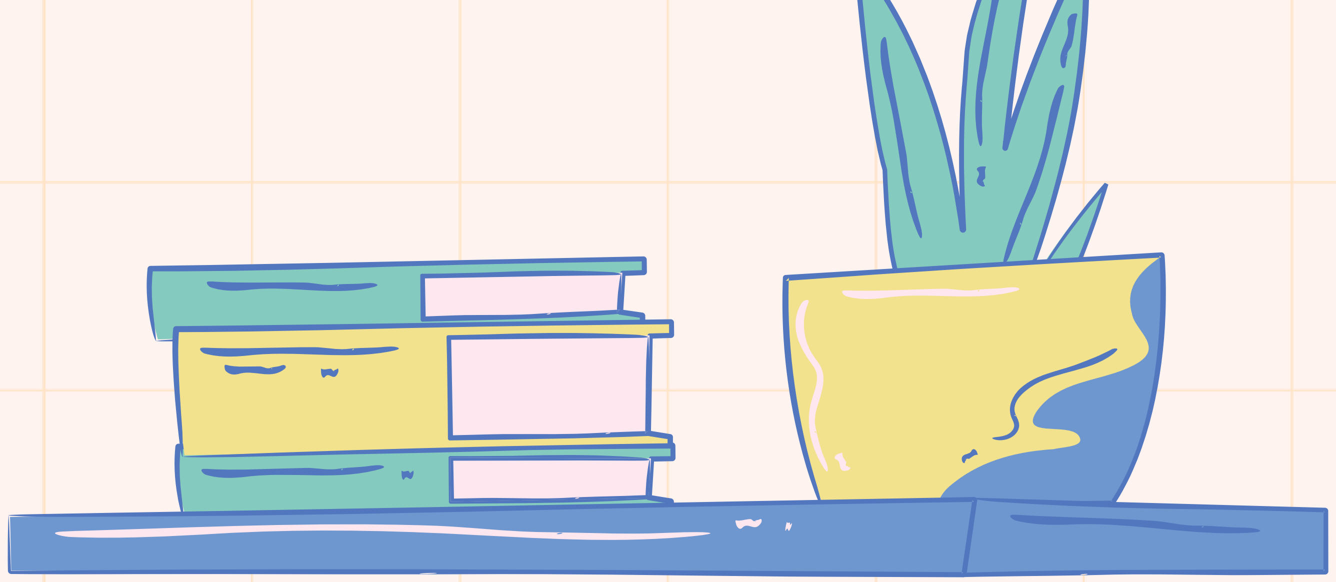
Figure 1: Effectiveness of $\mathrm{KD\text{-}LSTM}_{reg}$ vs. BERT_{base} and BERT_{large}



Dataset	LSTM$_{reg}$	BERT$_{base}$
Reuters	0.5 (1×)	30.3 (60×)
AAPD	0.3 (1×)	15.8 (50×)
IMDB	6.8 (1×)	243.6 (40×)
Yelp'14	20.6 (1×)	1829.9 (90×)

Table 3: Comparison of inference latencies (seconds) on validation sets with batch size 128.





5. CONCLUSION

Contributions:

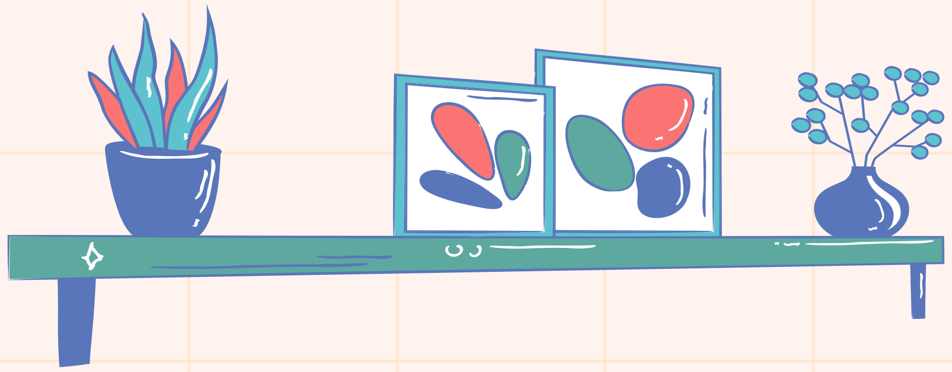
- Fine-tuning BERT to improve baseline models for document classification.
- Using knowledge learned by BERT to enhance the performance of a single-layer BiLSTM model (LSTMreg) through Knowledge Distillation.

Key Findings:

- After distillation, LSTMreg achieves performance comparable to BERTbase on most datasets.
- The resulting model:
  - Reduces the number of parameters by more than 30×
  - Achieves at least 40× faster inference

Future Work

- Investigating the impact of knowledge distillation across a wider range of neural network architectures.
- Developing model compression techniques specifically for Transformer models to reduce size and accelerate inference.



Thank You